

Forge — Implementation Plan

A hybrid-platform gym management application: **Owner dashboard on web, Trainer & Member apps on mobile** — with a shared design system and business logic.

1. Product Vision & Differentiation

Forge is designed to **feel light and intentional** — unlike bloated, enterprise-style gym software. Every screen serves a purpose, every animation gives feedback, and every metric provides an actionable insight. The core differentiators:

Differentiator	What It Means
Platform-Matched UX	Owner gets full-width dashboards on web; Trainers & Members get native mobile with camera, QR, and push notifications
QR Attendance	Frictionless check-in via a unique per-member QR code scanned at the gym entrance
Gamified Engagement	Streaks, badges, leaderboards, and monthly fitness challenges keep members coming back
Trainer Marketplace	Members can browse, compare, and book sessions with trainers — turning each trainer into a micro-brand
Hybrid Architecture	Shared business logic across web and mobile; Owner stays on web permanently, Trainer/Member graduate to native apps

2. Core Features & MVP Scope

2.1 Phase 1 — Web MVP (All Roles)

[!IMPORTANT] Phase 1 builds **all three modules as a web app** to validate every user flow before investing in native mobile. The Owner module stays web permanently; Trainer & Member modules serve as functional prototypes.

Authentication & Onboarding

- Role-based sign-up/login (Owner, Trainer, Member)
- Quick 3-step animated onboarding per role

Owner Module (Web — permanent home)

- Dashboard: active members, revenue, upcoming renewals, class utilization (charts)
- Member & Trainer directory with search/filter
- Payment & renewal analytics

- Push notification composer (mock)

Trainer Module (Web — prototype for Phase 2 mobile)

- Client list with profile cards
- Workout plan builder (exercise picker, sets/reps, schedule)
- Diet chart builder (meal plan grid) — *PT members only*
- Progress tracker (photo timeline + measurements log) — *PT members only*

Member Module (Web — prototype for Phase 2 mobile)

- Personal dashboard: today's workout, diet plan, streak counter, badges
- Workout log (mark exercises done)
- Calories burnt per exercise card (displays estimated calories for each logged exercise)
- Diet plan viewer — *PT members only*
- Attendance history + QR code for check-in
- Notification feed (renewals, trainer updates)

Novel Features (MVP-light versions)

- **Community Leaderboard:** Points from workouts logged, streaks, challenges
- **In-App Messaging:** Simple chat between trainer and member

2.2 Phase 2 — Native Mobile (Trainer & Member)

Trainer and Member modules graduate from web prototype to native mobile apps:

- **React Native apps** for iOS and Android
- Native camera access for progress photos
- Native QR code scanning for attendance
- Push notifications via Firebase FCM
- Offline-capable workout logging
- Bottom tab navigation optimized for one-handed use
- Owner module **remains on web** (no mobile port needed)

2.3 Phase 3 — Advanced Features

- AI Recommendation Engine — rule-based suggestions based on profile (age, weight, goal), displayed as "Forge AI Suggests"
- Body Composition Timeline — photo upload + manual measurements on a visual timeline
- Real ML-based recommendation engine (upgrade from rule-based)
- NFC attendance scanning
- Video exercise library with form tips
- Wearable device integration (Apple Watch, Fitbit)
- Owner financial forecasting / churn prediction
- Multi-branch gym management

- White-label / SaaS mode

3. Monetization Strategies

Strategy	Description
Freemium SaaS	Free for 1 trainer + 25 members; paid tiers for larger gyms
Trainer Marketplace Commission	10-15% commission on trainer session bookings
Premium Analytics	Advanced dashboards (churn prediction, revenue forecasting) for owners on Pro plan
In-App Purchases	Members buy premium workout programs, specialized diet plans
Affiliate Marketing	Supplement, gear, and equipment recommendations with affiliate links
White-Label Licensing	Gyms pay for branded versions of the app

4. Growth Hooks & Engagement

- **Streaks & Badges:** Daily workout streak, "Iron Week" (7 days), "Forge Master" (30 days)
- **Monthly Challenges:** Owner creates gym-wide challenges (e.g., "Most Workouts in March")
- **Referral Bonuses:** Members earn badge + points for referring friends
- **Progress Sharing:** Members can share body-comp timelines to social media (watermarked with Forge branding)
- **Trainer Spotlight:** Top-rated trainers featured on the community board
- **Smart Nudges:** "You're 1 workout away from your streak!" push notification

5. Tech Stack

Phase 1 — Web MVP (All Roles)

Layer	Choice	Why
Build Tool	Vite	Fastest HMR, excellent DX
UI	React 18	Component model, huge ecosystem
Routing	React Router v6	Standard, declarative routing
Styling	Vanilla CSS + CSS Variables	Full control, no framework overhead, shared design tokens
Charts	Recharts	Lightweight, React-native charting (Owner dashboards)
Icons	Lucide React	Clean, minimal icon set

Layer	Choice	Why
Animations	CSS transitions + Framer Motion (light)	Smooth micro-interactions
State	React Context + useReducer	Simple, no external dependency for MVP
Mock Data	JSON files + Context providers	Simulates backend; easy to swap for real API later

Phase 2 — Mobile Apps (Trainer & Member)

Layer	Choice	Why
Framework	React Native (Expo)	Cross-platform iOS + Android, shared JS knowledge
Navigation	React Navigation v6	Standard mobile navigation (stack, tabs, drawer)
UI Components	React Native Paper	Material Design components, dark theme support
Charts	Victory Native	Lightweight charting for mobile
Camera	expo-camera	Progress photos, document scanning
QR/Barcode	expo-barcode-scanner	QR check-in scanning
Push Notifications	Firebase FCM + expo-notifications	Reliable cross-platform push
State	Same Context + useReducer patterns	Consistency with web codebase
Offline	AsyncStorage + sync queue	Workout logs work without connectivity

Shared Across Platforms

Layer	Choice	Why
Business Logic	<code>packages/shared/</code> (TypeScript)	Calorie formulas, validators, data transforms — used by both web and mobile
API Client	Fetch / Axios wrapper	Same service layer, same request patterns
Design Tokens	JSON token file	Colors, spacing, typography shared between CSS variables and React Native StyleSheet
Types / Models	TypeScript interfaces	Single source of truth for data shapes

[!NOTE] Phase 1 is **frontend-only** with mock data. The architecture is designed so swapping mock data providers for real API calls is a one-line change per module. ~65% of non-UI code will be shared across

web and mobile.

6. UI/UX Design Language

Design Principles

- **Dark-first theme** with a warm accent palette (amber/orange on charcoal)
- **Card-based layouts** — every data unit lives in a softly elevated card
- **Generous whitespace** — the app should breathe
- **Mono + Sans pairing** — **Inter** for UI, **JetBrains Mono** for data/numbers
- **Micro-animations** — smooth page transitions, card hover lifts, progress bar fills, badge unlocks
- **Platform-adaptive**: full sidebar nav on web, bottom tab nav on mobile

Color Palette (Shared Design Tokens)

Token	Value	Usage
--bg-primary	#0F0F0F	App background
--bg-card	#1A1A1A	Card surfaces
--bg-card-hover	#242424	Card hover state (web) / pressed state (mobile)
--accent	#F59E0B	Primary accent (amber)
--accent-glow	#F59E0B33	Glow effects
--text-primary	#F5F5F5	Main text
--text-secondary	#A3A3A3	Secondary text
--success	#22C55E	Positive metrics
--danger	#EF4444	Alerts, overdue
--gradient-hero	linear-gradient(135deg, #F59E0B, #EF4444)	Hero elements

7. Project Structure

Phase 1 — Web MVP (Monorepo-ready)

```
d:\Forge Gym Management Application\  
├─ index.html  
├─ vite.config.js  
├─ package.json  
└─ public/
```

```

└─ forge-logo.svg
└─ src/
  └─ main.jsx          # Entry point
  └─ App.jsx           # Router + layout
  └─ index.css         # Global styles + design tokens
  └─ components/       # Shared UI components (web)
    └─ Sidebar.jsx
    └─ TopBar.jsx
    └─ StatCard.jsx
    └─ Badge.jsx
    └─ QRCode.jsx
    └─ ProgressRing.jsx
    └─ Modal.jsx
    └─ ChatBubble.jsx
  └─ layouts/
    └─ DashboardLayout.jsx # Sidebar + content shell
  └─ pages/
    └─ Login.jsx
    └─ Onboarding.jsx
    └─ owner/          # 🖥️ Permanent web home
      └─ OwnerDashboard.jsx
      └─ Members.jsx
      └─ Trainers.jsx
      └─ Revenue.jsx
      └─ Notifications.jsx
    └─ trainer/        # 📱 Web prototype → Mobile in Phase 2
      └─ TrainerDashboard.jsx
      └─ Clients.jsx
      └─ WorkoutBuilder.jsx
      └─ DietBuilder.jsx
      └─ ProgressTracker.jsx
    └─ member/         # 📱 Web prototype → Mobile in Phase 2
      └─ MemberDashboard.jsx
      └─ WorkoutLog.jsx
      └─ CaloriesBurnt.jsx
      └─ DietPlan.jsx   # PT members only
      └─ Attendance.jsx
      └─ Leaderboard.jsx
      └─ Messages.jsx
  └─ data/             # Mock data (JSON-like)
    └─ members.js
    └─ trainers.js
    └─ workouts.js
    └─ diets.js

```

```

├── analytics.js
├── context/
│   ├── AuthContext.jsx
│   └── AppContext.jsx
├── utils/
│   └── helpers.js                # Calorie formulas, formatters

```

Phase 2 — Monorepo After Mobile Addition

```

d:\Forge Gym Management Application\
├── packages/
│   ├── shared/                    # 🗑 Shared business logic
│   │   ├── package.json
│   │   ├── src/
│   │   │   ├── models/           # TypeScript interfaces
│   │   │   ├── services/         # API client layer
│   │   │   ├── utils/           # Calorie formulas, validators
│   │   │   └── constants/        # Design tokens as JSON
│   │   └── tsconfig.json
│   ├── web/                      # 💻 Owner Dashboard (React + Vite)
│   │   ├── index.html
│   │   ├── vite.config.js
│   │   ├── package.json
│   │   └── src/                  # Owner pages, components, styles
│   └── mobile/                   # 📱 Trainer + Member App (React Native)
│       ├── app.json
│       ├── package.json
│       ├── App.tsx
│       └── src/
│           ├── screens/
│           │   ├── trainer/       # Trainer screens
│           │   └── member/        # Member screens
│           ├── components/       # Mobile-specific components
│           ├── navigation/       # Stack + Tab navigators
│           └── theme/            # React Native StyleSheet from tokens
├── package.json                  # Monorepo root (workspaces)
└── README.md

```

8. Proposed Changes — Phase 1 (Current Build)

Project Initialization

[NEW] [package.json](file:///d:/Forge%20Gym%20Management%20Application/package.json)

Initialize via `npx create-vite@latest ./ --template react`. Adds Recharts and Lucide React as dependencies.

[NEW] [vite.config.js](file:///d:/Forge%20Gym%20Management%20Application/vite.config.js)

Standard Vite React config.

[NEW] [index.html](file:///d:/Forge%20Gym%20Management%20Application/index.html)

Entry HTML with Inter + JetBrains Mono font imports, meta tags, dark background.

Design System & Global Styles

[NEW] [index.css](file:///d:/Forge%20Gym%20Management%20Application/src/index.css)

All CSS custom properties (color palette, spacing, typography, shadows), base resets, card styles, button variants, animation keyframes, responsive breakpoints. These tokens are designed to map directly to React Native StyleSheet values in Phase 2.

Shared Components

[NEW] All files in `src/components/`

`Sidebar`, `TopBar`, `StatCard`, `Badge`, `QRCode`, `ProgressRing`, `Modal`, `ChatBubble` — reusable across all three role modules during Phase 1. In Phase 2, mobile-specific versions will be created in `packages/mobile/src/components/`.

Layouts

[NEW] [DashboardLayout.jsx]

(file:///d:/Forge%20Gym%20Management%20Application/src/layouts/DashboardLayout.jsx)

Sidebar + TopBar + scrollable content area. Collapses sidebar to bottom nav on mobile viewport (Phase 1 responsive fallback; replaced by native tab nav in Phase 2).

Pages — Auth

[NEW] [Login.jsx](file:///d:/Forge%20Gym%20Management%20Application/src/pages/Login.jsx)

Dark-themed login with role selector tabs, animated transitions. In Phase 2, Member/Trainer login moves to the mobile app.

[NEW] [Onboarding.jsx]**(file:///d:/Forge%20Gym%20Management%20Application/src/pages/Onboarding.jsx)**

3-step onboarding wizard with progress dots and slide animations.

Pages — Owner Module ( Web — Permanent)

[NEW] All files in `src/pages/owner/`

OwnerDashboard (KPI cards + charts), **Members** (searchable directory), **Trainers** (performance cards), **Revenue** (payment analytics), **Notifications** (composer UI). These pages stay in the web app permanently.

Pages — Trainer Module ( Web Prototype → Mobile in Phase 2)

[NEW] All files in `src/pages/trainer/`

TrainerDashboard (client overview), **Clients** (profile cards), **WorkoutBuilder** (drag-style exercise picker), **DietBuilder** (meal plan grid, PT members only), **ProgressTracker** (photo timeline + measurements, PT members only). These serve as functional prototypes; the UI logic and data flows will be re-implemented as React Native screens in Phase 2.

Pages — Member Module ( Web Prototype → Mobile in Phase 2)

[NEW] All files in `src/pages/member/`

MemberDashboard (today's plan + streaks + badges), **WorkoutLog** (checkbox exercise list), **CaloriesBurnt** (estimated calories per exercise card), **DietPlan** (meal cards, PT members only), **Attendance** (QR code + history), **Leaderboard** (ranked member list), **Messages** (chat with trainer). These serve as functional prototypes for Phase 2 migration.

Data & Context

[NEW] All files in `src/data/` and `src/context/`

Mock data sets for members, trainers, workouts, diets, analytics. Auth and App context providers. In Phase 2, mock data layers are replaced with API service calls from `packages/shared/`.

Utilities

[NEW] [helpers.js](file:///d:/Forge%20Gym%20Management%20Application/src/utls/helpers.js)

Shared utility functions: date formatting, calorie estimation formulas (MET-based), data transformers. In Phase 2, these move to `packages/shared/src/utls/`.

9. Verification Plan

Phase 1 — Web MVP (Browser-based)

1. **Start dev server:** `npm run dev` in the project root
2. **Login Flow:** Navigate to localhost, verify login page renders, select each role, confirm redirect to correct dashboard
3. **Owner Dashboard:** Verify KPI cards display data, charts render, member/trainer lists are populated
4. **Trainer Module:** Verify client list, workout builder interactions, diet builder grid, progress photo timeline
5. **Member Module:** Verify dashboard streaks/badges, workout log checkboxes, calories burnt card, QR code renders, leaderboard ranks, message bubbles
6. **Responsive:** Resize browser to mobile width, verify sidebar collapses to bottom nav, cards stack vertically (validates mobile-readiness for Phase 2)
7. **Animations:** Verify page transitions, card hover effects, and progress ring animations are smooth

Phase 2 — Mobile Apps (Device/Emulator)

- Run Trainer app on iOS simulator + Android emulator
- Run Member app on iOS simulator + Android emulator
- Verify native QR scanning works
- Verify push notifications are received
- Verify offline workout logging syncs when back online
- Verify camera captures progress photos

Manual Verification (User)

- Visually review the design and provide feedback on color palette, spacing, and overall feel
- Test Owner dashboard on desktop browser
- Test Trainer/Member apps on real physical devices